Available online at www.sciencedirect.com
ScienceDirect
Proceedings of the Combustion Institute 33 (2011) 301–307
Proceedings of the Combustion Institute
www.elsevier.com/locate/proci

An experimental and kinetic study of alkane autoignition at high pressures and intermediate temperatures

David J. Beerer, Vincent G. McDonell *

UCI Combustion Laboratory, Department of Mechanical and Aerospace Engineering, University of California Irvine, CA 92697-3550, USA

Available online 6 August 2010

Abstract

Autoignition delay times of individual alkanes and mixtures of alkanes were measured experimentally in a high pressure, intermediate temperature, turbulent flow reactor. The mixtures are intended to simulate various natural gas compositions from around the world. Testing was conducted from 7 to 15 atm and from 785 to 935 K which mimic the inlet conditions to the premixing zone in many modern lean premixed gas turbine engines. It is also in the kinetics region where reactions involving RO_2 and H_2O_2 control the ignition process. The results are compared to higher temperature autoignition data from the literature which reveals several interesting differences between the two regimes. Finally, the data are compared to simulations using comprehensive chemical kinetics mechanisms to elucidate the dominant reactions at these conditions.
© 2010 The Combustion Institute. Published by Elsevier Inc. All rights reserved.

Keywords: Autoignition; Ignition delay; Natural gas; High pressure; Flow reactor

1. Introduction

Autoignition is a spontaneous process whereby a mixture of fuel and air undergoes a chemical reaction leading to ignition and combustion without the aid of an external source such as a flame or spark. A common term associated with autoignition is the ignition delay time. This is the period of time between creation of the combustible mixture (e.g. injection of fuel into an airstream) and ignition. Ignition delay measurements are valuable for gas turbine engine design as well as for chemical kinetics research. Many modern gas turbine engines utilize a lean premixed combustion strategy where a premixing chamber upstream of the combustor mixes the fuel and air prior to combustion. Design engineers must ensure that the time spent by the fuel and air in the premixer is less than the ignition delay time in order to prevent autoignition. Typical inlet conditions to the premixer are between 5 and 30 atm and temperatures from 600 to 1000 K [1]. For kinetics research, the ignition delay time represents one type of experimental data which is useful in validating chemical kinetics mechanisms. In both applications, accurate knowledge of ignition delay as a function of pressure, temperature, equivalence ratio and fuel composition is required. Therefore, the goal of the present study is to quantify experimentally the ignition

* Corresponding author. Fax: +1 949 824 7423.
E-mail address: mcdonell@ucicl.uci.edu (V.G. McDonell).

1540-7489/$ - see front matter © 2010 The Combustion Institute. Published by Elsevier Inc. All rights reserved.
doi:10.1016/j.proci.2010.05.015

delay times of several fuels at these elevated pressure and temperature conditions. The fuels of interest in this paper are alkane blends which represent various natural gas compositions from around the world, especially those with larger fractions of ethane and propane [2].

2. Background

The most common experimental devices for ignition delay measurements are shock tubes, rapid compression machines (RCMs) and flow reactors. Shock tubes work well at high temperatures ($T > 1200$ K) because of their ability to rapidly bring their mixture to test conditions. But they are limited to test times on the order of a few milliseconds, so are not capable of testing at lower temperatures [3]. RCMs have been widely used in the low to intermediate temperature region ($1100\ \text{K} > T > 700\ \text{K}$) and have garnered valuable ignition delay data [4]. One difficulty with RCMs is the inevitable heat loss to the walls which limits test times to typically around a 100 ms. Flow reactors mimic the flow conditions inside gas turbine premixers so their data has been used for both gas turbine design [5] as well as chemical kinetics research [6]. In order to obtain useful kinetics data from flow reactor experiments, many of the secondary effects such as mixing time, heat loss and diffusion of species must be rendered negligible in order to model the experiment as a plug flow. Heater and facility capabilities limit experiments to temperatures of around 1000 K and pressures from 1 to 30 atm.

One of the first comprehensive studies relevant to natural gas autoignition was that of Burcat et al. [7] who measured ignition delay times of alkanes C_1 through C_5 in a shock tube at temperatures between 1200 and 1900 K. Burcat et al.'s [7] measurements found methane to have the longest ignition delay of any alkane. Propane, butane and pentane all had similar ignition delays times which were between 5 and 10 times shorter than methane depending on the temperature. Somewhat peculiarly, ethane's ignition delay was shortest, being roughly half that of the heavier alkanes at these conditions, the reason for which will be discussed later. The measured overall activation energies of ethane through pentane ranged from 155 to 167 kJ/mol, while methane's activation energy was 209 kJ/mol. Crossley et al. [8] continued this work with shock tube measurements of mixtures of methane with fractions of alkanes, ethane through pentane, in quantities of up to 20%. Eubank et al. [9] along with Spadaccini and Colket [10] also independently studied methane/higher alkane mixtures in shock tubes. The temperatures for all these experiments ranged from 1200 to 1800 K with pressures between 1 and 5 atm. Ignition delay times ranged from 15 μs to 2 ms. Westbrook and Pitz [11,12] have developed several comprehensive chemical mechanism for high temperature natural gas ignition. Simulations with their mechanisms can reproduce the experimental data of Burcat et al. [7] and have identified many of the controlling chemical reactions and species leading to ignition at high temperatures.

Recently, the research groups of Petersen and Curran [4,13] together have measured ignition delay times of alkane mixtures in shock tubes and RCMs for both high and intermediate temperatures. Conditions in the RCM ranged from 10 to 40 atm and from 700 to 1000 K with measured delay times between 5 and 300 ms. Shock tube testing have been conducted between 1 and 40 atm and temperatures from 1100 to 1600 K with measured delay times between 20 μs and 2 ms. They have also developed several comprehensive kinetics mechanisms that can accurately reproduce their experimental findings at both high and intermediate temperatures.

Ignition delay times have also been measured quite extensively with flow reactors in past decades. Lezberg [14] measured delay times of propane at atmospheric pressure between 1030 and 1135 K, while Lefebvre et al. [15,16] measured delay times of methane and propane at pressures between 1 and 10 atm and temperatures from 800 to 1100 K. Ignition delay measurements of binary and ternary mixtures of alkanes have been studied recently at atmospheric pressure by Holton et al. [17] at temperatures between 930 and 1100 K. Measured delay times in these studies ranged from 20 to 200 ms.

The primary goal of the current study is to quantify the effects of adding higher alkanes to methane at intermediate temperatures and pressures up to 15 atm. This is accomplished in a high pressure, turbulent flow reactor with residence times of up to 700 ms which allows for ignition delay measurements at conditions outside the capabilities of other comparable flow reactor, RCM, or shock tube experiments.

3. Experiment

The flow reactor used in the present study is designed to measure ignition delay times of gaseous fuels inside a turbulent air flow at pressures and temperatures up to 15 atm and 950 K, respectively. A more detailed description of the flow reactor can be found elsewhere [18,19] however a short summary is presented here.

The flow reactor test section consists of a stainless steel cylindrical duct with an inner diameter of 38 mm and a length of 3.90 m. Fuel is injected by way of eight radially inward pointing jets (0.8 mm diameter) in the converging section of a venturi at the beginning of the test section and mixed with the oncoming air. The fuel/air mixture flows down

the duct until ignition occurs some distance downstream. An annotated schematic of the flow reactor is shown in Fig. 1. Ignition sensing equipment (e.g. thermocouples, pressure transducers, photodiodes, and photomultiplier tubes) are located along the duct to determine the time and position of autoignition. The pressure, temperature, flow rates, and fuel composition can all be independently controlled and set to any desired value so that a parametric study can be conducted. Air and fuel flow rates and temperatures are measured prior to injection and an energy balance is applied to calculate the mixture temperature. Wall heaters surround the vessel to minimize heat loss during each test; however a small axial temperature gradient does exist. The uncertainty of the reported mixture temperatures and pressure for each ignition event is estimated to be around 10 K and 0.2 atm, respectively which result from small fluctuations in the heater output and flow meter controllers.

At the start of each test, a fast acting solenoid valve is activated which directs fuel into the fuel line and into the reactor. A Helium Neon laser is used to detect the instant fuel enters the test section, $t_{\text{injection}}$. The laser emits an infrared beam with a wavelength 3.39 μm which passes through the vessel perpendicular to the flow by way of two quartz windows located in the optical ports just after the venturi section as seen Fig. 1. This signal is received by a thermoelectrically cooled photo-detector. The absorbance of hydrocarbon fuels at this wavelengths is quite large [20] so the instant of fuel injection is seen as a rapid attenuation of the laser signal. The fuel concentration can also be calculated through the relation between the initial and attenuated beam intensity using Beer's law.

A metal clad fiber optic probe is used to determine the time of autoignition, t_{ignition}. The probe is located at the end of the test section and has a line of sight down axis of the flow reactor. It detects light emission from the ignition event regardless of where the mixture ignites. The probe is connected to a Hamamatsu R106 type photomultiplier tube (PMT) which observes emissions over its full operational spectrum. Thermocouples are used to measure the temperature of the fuel and air (prior to mixing) and at the end of the test section. No thermocouples are located immediately after the venturi so as to not disturb the flow. Pressure transducers are located at the beginning and end of the test section to measure the reactor's nominal pressure and any pressure rise after ignition. Diagnostic data is fed into two computers using National Instruments Labview® software.

The ignition delay time, τ, is measured as the difference in time between fuel injection into the test section (as detected by the laser) and light emission (as detected from the PMT):

$$\tau = t_{\text{ignition}} - t_{\text{injection}}$$

Figure 2 shows a recording for one particular ignition delay test. Fuel entered the flow reactor at $t = 0.072$ s into the run and autoignition was detected at $t = 0.324$ s. The time difference between two events was 0.252 s, which is the ignition delay time. Autoignition is also accompanied by a short pressure spike of around 70 kPa lasting about 0.1 s and a rise in temperature. After ignition, the fuel flow is terminated and the rig is allowed to settle back to pre-ignition conditions before further testing. The repeatability of any measurement is usually around 15% of the total ignition delay time. This is mostly due to the difficulty in maintaining identical testing conditions throughout the duration of the experiment, for any small perturbation in the inlet temperature, pressure or mixing can affect the overall ignition delay time. The relatively uncertainty for the injection and ignition measurement is only a few milliseconds due to the inherent response time of the either photo-detector or PMT, but this uncertainty is contained within the usual scatter of the data.

Table S1 in Supplementary material lists the composition of all the test fuels in this study. A

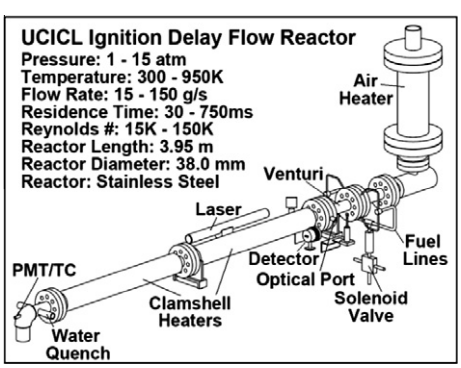

Fig. 1. UC Irvine Combustion Lab flow reactor.

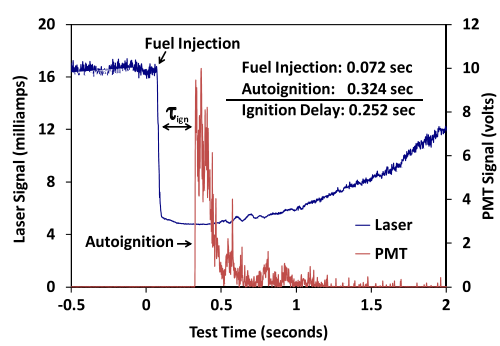

Fig. 2. Laser and PMT recording from one ignition delay test (methane/propane – 90/10 blend, $T_{\text{mix}} = 862$ K, $P = 9.0$ atm, $\varphi \sim 0.6$, $\tau = 252$ ms).

total of 12 different fuels were examined. Fuels 1–3 consisted of individual alkanes: methane, ethane and propane. Fuels 4–6 consisted of binary blends of methane with up to 30% ethane by volume. Fuels 7–9 consisted of binary blends of methane with up to 30% propane by volume. Fuels 10–12 consisted of blends of three or more alkanes including small quantities of species up to C_6 along with some diluents (e.g. nitrogen and carbon dioxide). In fact, Fuel 12 is natural gas taken from the local pipeline at the laboratory in Irvine, California. The fuel purity of the individual methane, ethane and propane tested were 99.99%, 99.0% and 99.99%, respectively. Percentages of higher alkanes in the mixtures with methane were within 1% of the stated value. Testing was conducted at pressures between 7 and 15 atm and mixture temperatures from 785 to 935 K with equivalence ratios around 0.4–0.6 and nominal air flow rates from 34 to 50 g/s. The majority of the testing was conducted at a standard condition of 9 atm, equivalence ratio of 0.6 and an air flow rate of 34 g/s. By maintaining identical test conditions inside the rig for each fuel, the effect of the chemical composition on the ignition delay time can be isolated and studied.

4. Results

The results and test conditions for all 997 autoignition events measured in this study are tabulated in Tables S2–S15 in Supplementary material. The ignition delay times of individual methane, ethane and propane are presented in Fig. 3 in the context of previous flow reactor measurements found in the literature [14–17]. The ordinate axis in this Arrhenius plot is the product of the delay time and the pressure which normalizes the various sets of data obtained at different pressures. This inverse pressure dependency assumption will later be shown to hold reasonably well for alkane mixtures at these conditions. At more intermediate temperature conditions methane's ignition delay times are two to three times longer than ethane or propane. Ethane's ignition delay times are now in between methane and propane's delay times, and propane's delay times are shortest of the three fuels. This same qualitative observation can also be made with previous literature data at intermediate temperatures. Ignition delays from this study ranged from 60 ms at the highest temperatures to as long as 600 ms at the lower temperatures. The overall activation energies of methane, ethane and propane were found to be 77, 140, and 125 kJ/mol, respectively (with an estimated uncertainty for each around ± 10 kJ/mol). These values are less than those measured by Lefebvre et al.'s [15,16], who obtained 105 and 160 kJ/mol for methane and propane, respectively, at slightly higher temperatures than in this study. Lefebvre's [15,16] values are themselves less than Burcat's [7] measured activation energies obtained at even higher temperatures. Hence, the overall activation energy appears to increase with increasing temperature.

The ignition delay times of binary mixtures of methane/ethane and methane/propane are presented in Figs. 4 and 5, respectively. Two features observed for both sets of data were a monotonic

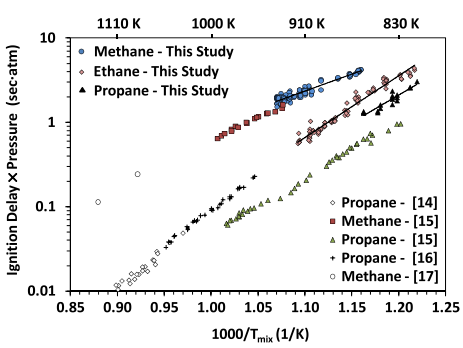

Fig. 3. Ignition delay times of methane, ethane and propane at 9 atm and $\varphi \sim 0.6$ from this study compared to delay times from other flow reactor investigations.

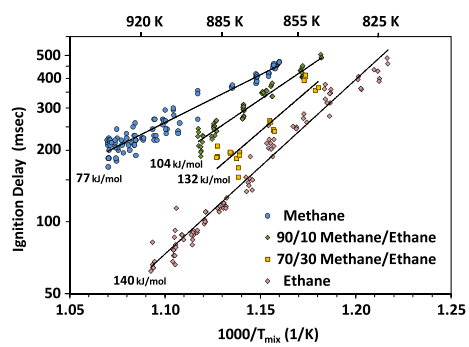

Fig. 4. Ignition delay of binary blends of methane and ethane at 9 atm and $\varphi \sim 0.6$.

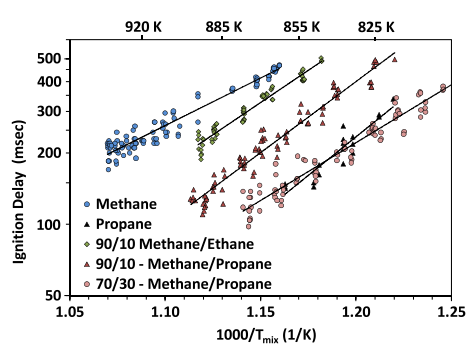

Fig. 5. Ignition delay of binary blends of methane and propane or ethane at 9 atm and $\varphi \sim 0.6$.

decrease in ignition delay time with increasing ethane or propane fraction, and a monotonic increase in overall activation energy with increasing ethane or propane fraction. It is interesting to note that, in Fig. 5 when the propane fraction approaches 30% the delay time is, to within the experiments uncertainty, identical to that of pure propane. While the trends for mixtures of methane with ethane or propane are similar, quantitatively propane reduces the delay time more so than ethane for the same fraction in the fuel. This is apparent in Fig. 5 where the ignition delays for the blend with 10% propane are notably shorter than the blend with 10% ethane. Fuels with alkanes C_4 through C_6 of around 0.1–1% (with ethane and propane fractions from 5 to 10%) were tested but are not shown. The delay times of these fuels were roughly 1.5 times shorter than those of pure methane at similar test conditions. Hence, the effects on the ignition delay time from these alkanes even in trace quantities are indeed not negligible.

Several measurements were made to asses the impact of pressure on the ignition delay time. Figure 6 presents the ignition delay times of a methane/ethane/propane mixture at pressures of 7, 9 and 15 atm. Overall, the ignition delay time decreases monotonically with increasing pressure. The delay time was found to scale with a $P^{-1.0 \pm 0.1}$ dependency, where P is the absolute pressure. The overall activation energy also decreases very slightly with increasing pressure. While not shown, measurements from other alkane blends displayed similar pressure dependencies. This inverse pressure relation justifies the ordinate axis used in Fig. 3 to normalize the data sets measured at different pressures.

Finally, a brief experimental sensitivity analysis was performed on the effects of flow conditions within the test section on the ignition delay times in this study. Overall the measured delay times were found to be unaffected by varying the equivalence ratio between 0.4 and 0.6 or air flow rates from 34 to 51 g/s for any of the fuels studied. No direct studies were performed to look at the impact of fuel and air mixing on delay time, however changing either fuel or air flow rates will alter the degree of fuel jet penetration into the air stream and hence the mixing rate; yet the ignition delay appeared to be independent of these factors, which suggest that their impact on the delay time are small (at least to within the range we tested for hydrocarbon fuels). This analysis is important to consider because in recent years some debate has occurred on the ability to measure ignition delays at intermediate temperatures under near ideal condition. This is especially so with regard to shock tubes experiments with hydrogen and even hydrocarbons [3,21]. While it is difficult to say with complete certainly that the experimental results in this study are only a result of chemistry and not other experimental artifacts, the results at minimum are accurate on a relative basis as almost all the measurements were made at nearly identical test conditions within the flow reactor.

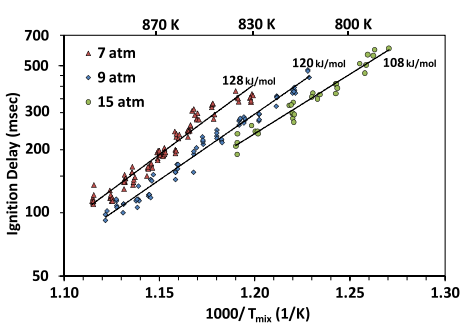

Fig. 6. Pressure effect on ignition delay of the 70/15/15 methane/ethane/propane fuel blend with $\varphi \sim 0.6$.

5. Discussion

The results presented in the previous section are now compared to higher temperature data which reveals some interesting differences in the autoignition process in the two temperature regimes. For one, while in both regimes methane's delay time is the longest of all the alkanes, it is much more so at higher temperatures. Burcat et al. [7] observed methane's delay time to be five to ten times that of the other alkanes, while this study indicates a factor of only two to three. A second major contrast between the two regimes is the relative reactivity of ethane and propane. Burcat et al. [7] observed ethane's delay time to be shorter than that of propane, whereas this study observed the opposite. Third, Burcat et al. [7] observed that the overall activation energy of methane is 40 kJ/mol larger than the other alkanes at high temperatures; whereas at intermediate temperatures, it is about 40 kJ/mol less. These differences are attributed to the chemical kinetics responsible for ignition in the two temperature regions, and not to the different experimental techniques.

The ignition delay trends of alkane mixtures between high and intermediate temperatures are also quantitatively different in many ways. The present study indicates a reduction in delay time with increasing fractions of higher alkanes in the fuel at intermediate temperatures, but the relative reduction is not as great compared to higher temperatures. For example, Eubank [9] reports that blends of methane/ethane (83/17) and methane/ethane/propane (77/15/8) reduce the delay time in comparison to pure methane by a factor of 3 and 9, respectively, at temperatures from 1300 to 1600 K. The present results indicate a reduction in delay time relative to pure methane for a methane/ethane blend (85/15) and methane/ethane/

propane blend (70/15/15) by only a factor of 1.2 and 2.7 at 875 K, respectively. The greater sensitivity at higher temperatures is attributed to the larger disparity between the methane's delay times to other alkanes.

To further illustrate this point, Fig. 7 displays the ignition delay times for binary methane/ethane blends as a function of ethane fraction for several temperatures. The curves for 850 and 950 K were obtained through interpolation of the data from this study. The curves for 1400, 1500 and 1600 K were taken from results of the kinetic simulation by Westbrook [11]. Notice the relative reduction in delay time with increasing ethane fraction at intermediate temperatures (especially 850 K) is less compared to the mixtures at 1400–1600 K. It is interesting to note though that both temperature regimes are most sensitive to ethane fractions around 0–10%. The ignition delay continues to reduce with increasing ethane content beyond 10%, but at a diminishing rate. These trends also hold for binary mixtures of methane and propane.

Eubank et al. [9] and Spadaccini and Colket [10] also noted with alkane mixtures, a constancy of activation energy with increasing higher alkane fraction in the fuel. However the present work (Figs. 4 and 5), indicates that overall activation energy is dependent on the fraction of higher alkanes in the mixture. This makes developing a general correlation for the current data rather difficult. For instance, Spadaccini and Colket [10] were able to correlate all their data into an Arrhenius expression in the form $\tau = A \exp(E/RT)\,[O2]^a[CH4]^b[HC]^c$ where A and E are constant value pre-exponential term and activation energy, respectively, and the effect of adding higher hydrocarbons is accounted for by the inclusion of the hydrocarbon concentration term [HC]. Because of the complex relation between fuel compositions and ignition delay at intermediate temperatures, the current data could not be fit to such a simple expression.

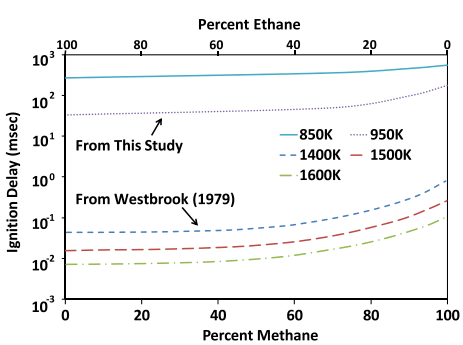

Fig. 7. Ignition delay of binary methane/ethane mixtures as a function of ethane fraction for a density of 2.75×10^{-3} gm/cm^3. Westbrook's (1979) curves are from simulations, where as this study's curves are interpolated from experimental data.

The general chemical kinetics responsible for autoignition at high, intermediate and low temperature are summarized by Westbrook [22]. He states that above 1200 K alkyl (R) radicals formed through H-abstraction reactions undergo β-scission reactions to form olefins along with H and/or CH_3 radicals. The ignition delay time is dependent on the ratio of CH_3 to H radicals formed during the β-scission reactions. Pure methane forms only slow reacting CH_3 radicals leading to long ignition delays, while ethane only forms highly active H atoms resulting in short ignition delays. C_{3+} alkanes generate a mixture of CH_3 and H radicals which leads to ignition delay times in between that of methane and ethane, which explains the observations made by Burcat et al. [7]. Addition of higher alkanes to methane accelerates the autoignition process by introducing reactive H radicals into the system. In the intermediate temperature region (1200 K > T > 850 K), the rate of formation and decomposition of hydrogen peroxide, H_2O_2, controls the length of the ignition delay time [22]. Below 850 K, the formation and oxidation of alkyl-peroxy (RO_2) radicals control the rate of ignition. The present study's temperature range (785–935 K) falls into the transition region between the RO_2 and H_2O_2 controlled ignition.

A simple modeling effort was undertaken to see if the observations from this study could be reproduced using kinetic simulations with two mechanisms which include both RO_2 and H_2O_2 chemistry [13,23]. Simulations were performed in CD-Adapco's DARS® software using a single homogenous plug flow reactor with constant enthalpy and pressure. Figure 8 presents a comparison of some of the simulations (solid curves) using the mechanism from the Galway Combustion Chemistry Centre [13] without any rate constant modification to the experimental data (points). The simulations were, on average, within 20% of the measured values and exhibit the proper intermediate temperature trends in terms of overall activation energy, relative fuel reactivity and

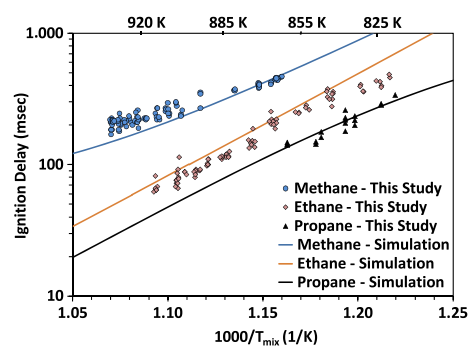

Fig. 8. Comparison between experimental data from this study and an adiabatic homogenous kinetic simulation using the mechanism from the Galway Combustion Chemistry Centre.

pressure effect. Simulations using the mechanism of Ribaucour et al. [23] were on average within 25% of the experimental data. While not shown, the model also accurately reproduced the effects of adding ethane and propane to the fuel as well.

Species flux analysis from the software indicated that the ignition process at intermediate temperatures is controlled by the rate of formation of H_2O_2. The hydrogen peroxide then decomposes into active OH radicals, increasing the reactivity of the system. At these elevated pressures and intermediate temperatures, atomic hydrogen reacts with molecular oxygen and a third body via the reaction: $H + O_2 + M \rightarrow HO_2 + M$ to form hydroperoxy. The HO_2 radical then attacks the alkanes via the reactions: $RH + HO_2 \rightarrow R + H_2O_2$, where R is the methyl, ethyl or propyl radical. The latter reaction is the primary source of H_2O_2 at these conditions. At 850 K, the relative rates of the three reactions are from fastest to slowest: $C_3H_8 + HO_2 \rightarrow C_3H_7 + H_2O_2$, $C_2H_6 + HO_2 \rightarrow C_2H_5 + H_2O_2$, $CH_4 + HO_2 \rightarrow CH_3 + H_2O_2$. Hence adding either ethane or propane to methane will lead to a higher formation rate of H_2O_2, resulting in a shorter ignition delay time. Because propane's recombination reaction rate is faster than ethane's, propane is more effective at reducing the ignition delay at intermediate temperatures, which is consistent with the experimental findings.

6. Conclusions

Ignition delay times were measured in a flow reactor at pressures from 7 to 15 atm and temperatures from 785 to 935 K. Fuels tested include individual alkanes and mixtures of alkanes that represent various natural gas compositions. A number of differences in ignition delay trends were identified between high and intermediate temperatures, including the overall activation energies, relative reactivity of ethane and propane, and impact of small quantities of ethane or propane mixed with methane on the ignition delay time. These contrasting trends are attributed to the different elementary reactions that control the ignition process in the two temperature regimes. Whereas at higher temperatures, ethane and propane addition accelerates the autoignition through an increase in the rate of formation of H radicals; at intermediate temperature, ethane and propane addition augment the rate of formation H_2O_2 which ultimately leads to a reduction in the ignition delay time.

Acknowledgments

The authors would like to thank Prof. Scott Samuelsen, Richard Hack, James Guo, James Hu, Patrick Randall and David Page at the UCICL. Insightful discussions with Drs. Petersen, Westbrook, Dryer, Santoro and Colket are acknowledged. Thanks are also given to EPRI for financial assistance under contract EP-P28862/C13652 and the U.S. Department of Energy's University Turbine Systems Research program.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.015.

References

[1] T. Lieuwen, V. McDonell, D. Santavicca, T. Sattelmayer, *Combust. Sci. Technol.* 180 (6) (2008) 1169–1192.
[2] W. Liss, W. Thrasher, G. Steinmetz, P. Chowdiah, A. Attari, *GRI*-92/1023, 1992.
[3] C. Aul, E. Petersen, 6th U.S. Nat. Combust. Meeting 12-F-5, 2009.
[4] D. Healy, H. Curran, S. Dooley, et al., *Combust. Flame* 155 (2008) 451–461.
[5] L. Spadaccini, J. TeVelde, *Combust. Flame* 46 (1982) 283–300.
[6] P. Gokulakrishnan, G. Gaines, J. Currano, M. Klassen, R. Roby, *J. Eng. Gas Turb. Power* 129 (2007) 655–663.
[7] A. Burcat, K. Scheller, A. Lifshitz, *Combust. Flame* 16 (1) (1971) 29–33.
[8] R. Crossley, E. Dorko, K. Scheller, A. Burcat, *Combust. Flame* 19 (3) (1972) 373–378.
[9] C. Eubank, M. Rabinowitz, W. Gardiner, R. Zellner, *Proc. Combust. Inst.* 18 (1981) 1767–1774.
[10] L. Spadaccini, M. Colket, *Prog. Energy Combust. Sci.* 20 (1994) 431–460.
[11] C. Westbrook, *Combust. Sci. Technol.* 20 (1) (1979) 5–17.
[12] C. Westbrook, W. Pitz, *Combust. Sci. Technol.* 33 (5) (1983) 315–319.
[13] E. Petersen, D. Kalitan, S. Simmons, G. Bourque, H. Curran, J. Simmie, *Proc. Combust. Inst.* 31 (2007) 447–454.
[14] E. Lezberg, NASA TN-4028, 1957.
[15] A. Lefebvre, W. Freeman, L. Cowell, NASA CR-1750, 1986.
[16] G. Freeman, A. Lefebvre, *Combust. Flame* 58 (1984) 153–162.
[17] M. Holton, M.P. Gokulakrishnan, M. Klassen, R. Roby, G. Jackson, *Proc. ASME Turbo Expo*, Orland, Florida, 2009, GT 2009-59309.
[18] D. Beerer, *Autoignition of Methane, Ethane, Propane and Hydrogen in Turbulent High Pressure and Temperature Flows*, MS thesis, University of California, Irvine, USA, 2009.
[19] D. Beerer, V. McDonell, S. Samuelsen, S.L. Angello, *Proc. ASME Gas Turbo Expo*, Orlando, Florida, 2009, GT 2009-60268.
[20] D. Burch, D.B. Edwards, *J. Opt. Soc. Am.* 55 (2) (1964) 174–177.
[21] M. Chaos, F. Dryer, *Int. J. Chem. Kinet.* 42 (3) (2010) 143–150.
[22] C. Westbrook, *Proc. Combust. Inst.* 28 (2000) 1563–1577.
[23] M. Ribaucour, R. Minetti, L. Sochet, H. Curran, W. Pitz, C. Westbrook, *Proc. Combust. Inst.* 28 (2000) 1671–1678.